



QC-Stark: A Multi-Task Benchmark Revealing LLM Capability Dissociations in Quantum Computing

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2,750 evaluations · 10 models · 11 tasks · 5 difficulty levels · 5 seeds · auto-verified via execution



Scan Me!

Motivation

Existing quantum computing (QC) benchmarks cover only slices of the practitioner workflow—contest programming, multiple-choice Q&A, or educational exercises. No prior work evaluates the **full operational spectrum**—construction, debugging, compilation, error correction, and simulation—with systematic difficulty scaling and psychometric validation. **The core problem:** LLM capabilities are not unidimensional. A model ranked **#1 overall can score 0% on debugging**. Single-score leaderboards hide task-specific failures that matter for deployment.

Four Design Principles of QC-Stark

- 11 tasks** spanning the full quantum computing workflow: circuit construction, debugging, compilation, verification, simulation, and error correction.
- 5 difficulty levels** per task—TEXTBOOK → HOMEWORK → EXAM → RESEARCH → OPEN—for fine-grained capability diagnosis.
- Seed-based procedural generation:** each instance is generated from a (task, level, seed) triple—contamination-resistant, no static test set to memorise.
- Automatic verification** via Qiskit execution—state fidelity, functional equivalence, syndrome correctness; no human grading required.

Related Work

Benchmark	Multi-task	Diff. levels	Auto-verify	IRT	Best acc.
Quantum-Audit '26	○	○	○	○	84%
QuantumKatas '26	~	○	✓	○	83%
QCoder '25	○	○	✓	○	78%
QCircuitBench '24	~	○	✓	○	—
SciCode '24	✓	○	✓	○	4.6%
QC-Stark (ours)	✓	✓	✓	✓	55.6%

✓ = yes ~ = partial ○ = no

IRT: Item Response Theory

Task Coverage: 11 Tasks × 5 Levels × 5 Seeds

T#	Task	Stage	Mean acc.
T1	State Preparation	Construct	0.440
T2	Trotterization	Construct	0.252
T3	Oracle Synthesis	Construct	0.412
T4	Debugging	Understand	0.020
T5	Noise Discrimination	Understand	0.272
T6	Reverse Engineering	Understand	0.448
T7	Equivalence Checking	Verify	0.824
T8	Hardware Routing	Compile	0.184
T9	Noise Fidelity	Simulate	0.516
T10	VQE	Simulate	0.592
T11	Quantum Error Correction Decoding	Error Correction	0.440

Difficulty span: $41.2\times$ ($T4 = 0.020 \rightarrow T7 = 0.824$). All 5 seeds per (task, level) pair; 25 instances per task.

Key Takeaways

- Capability dissociation is real and large.** Four of five top-ranked models score 0% on debugging (T4) while last-ranked LLaMA-70B outperforms them. Gemini FL (#7 overall) leads hardware routing (T8). Single-score leaderboards cannot capture this.
- Difficulty scaling reveals sharp cliffs.** Mean accuracy drops 62% from Level 1 to Level 5; hardware routing collapses to 0.07 at Level 3+. IRT marginal reliability $r = 0.984$, confirming psychometric validity.
- Rankings robust to prompt format** ($\rho = 0.95, p < 0.001$); absolute scores shift up to +0.215 on T11.

Model Rankings (Mean Accuracy $\pm \sigma$; 275 instances each)

Rank	Model	Overall	T4	T8	T7	T11	IRT θ
1	o4-mini	.556 $\pm$.045	.00	.16	.92	.52	+0.649
2	Sonnet 5	.553 $\pm$.068	.00	.12	1.00	.68	+0.410
3	GPT-5.4	.545 $\pm$.055	.16	.08	.96	.72	+0.842
4	Gemma-4	.491 $\pm$.088	.00	.28	.76	.64	+0.304
5	Opus 4.1	.396 $\pm$.035	.00	.12	.96	.40	−0.455
6	GPT-4.1m	.371 $\pm$.055	.00	.16	1.00	.32	−0.362
7	Gemini FL	.367 $\pm$.067	.00	.44	1.00	.08	−0.454
8	Gemini 3.5F	.309 $\pm$.023	.00	.16	.64	.20	−1.025
9	Mistral L3	.291 $\pm$.023	.00	.24	.96	.48	−0.976
10	LLaMA-70B	.120 $\pm$.027	.04	.08	.04	.36	−2.148
Task mean		.400	.02	.18	.82	.44	

σ = std dev across 5 seeds. **T4**=Debugging, **T8**=Routing, **T7**=Equiv., **T11**=QEC. **Red** = paradox cell.

Three Paradoxes in LLM Behavior

- Debugging (T4): rank-1 models produce 0% working code** ($\rho = +0.50, p = 0.14$). Four of five top-ranked models score 0%; LLaMA-70B (#10) outperforms them at 4%. Correct responses use 1,111–5,056 fewer tokens than wrong ones—across all 10 models.
- Routing (T8): orthogonal to all simulation tasks** ($r = -0.41$ vs. Trotterization). Gemini FL (#7 overall) leads at 0.44. Mean at Level 3+: **0.011** across all models (universal cliff).
- Reasoning effort (T11/QEC): more compute degrades performance.** GPT-5.4 at low reasoning effort: 100% correct; at high effort: accuracy falls to 44% while crash rate rises to 56%. o4-mini at low effort: 51% false positives (over-corrects error-free syndromes); at high effort: crash rate rises to 16%. The two models degrade in structurally opposite directions.

Behavioral Analysis: Evidence from 2,750 Outputs

Correct responses are shorter (all 10 models, no exceptions): Mean token counts for correct vs. incorrect responses, verified across the full dataset:

Model	Mean tokens (correct)	Mean tokens (wrong)	Difference
Sonnet 5	3,180	8,236	−5,056
o4-mini	3,127	5,078	−1,952
GPT-5.4	1,263	2,374	−1,111

Debugging anatomy (T4): Sonnet-5 (#2 overall) produces **zero executable code** across all 25 instances—mean 13,790 tokens of prose with no `solve()` function returned. GPT-5.4 (#3) passes 5/25 with mean 1,870 tokens of direct gate-level fixes.

Noise Discrimination (T5) — training data recency boundary: Three models fail all 25 instances via deprecated `import qiskit.providers.aer.noise` (removed in Qiskit 1.0, 2024). Top-5 models use the current `DensityMatrix` interface and score 0.70–0.89—a training data recency boundary, not a QC reasoning gap.

Item Response Theory (IRT) Validation [2-PL, 275 Items]

Statistic	Value	Statistic	Value
Reliability	$r = 0.984$	Items $a > 1.0$	77.8%
Ability span	3.0 s.u.	Items $a > 2.0$	43.6%
GPT-5.4 (IRT #1) $\theta = +0.842$		o4-mini (acc. #1) $\theta = +0.649$	

GPT-5.4 leads IRT ($\theta = +0.842$) but ranks #3 in accuracy: IRT binarises at score > 0.5 . Difficulty monotonically spans ability range (L1: $b = -0.89 \rightarrow$ L5: $b = +0.68$). Rankings robust to prompt format ($\rho = 0.95, p < 0.001$); max shift +0.215 on T11.

References

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Code & data: github.com/prannerta100/qc-stark